

SQL QUERIES

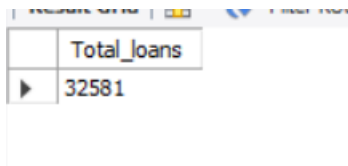
CREDIT RISK AND LOAN DEFAULT ANALYSIS

Business Problem:

How can a lending institution identify high-risk loan segments, understand the key factors associated with loan defaults, and reduce credit risk across its loan portfolio?

Inspection of data:

```
SELECT COUNT(*) AS Total_loans  
FROM credit_risk;
```



Total_loans
32581

Checking null / missing values:

```
SELECT  
    SUM(Person_age IS NULL) AS null_age,  
    SUM(Person_income IS NULL) AS null_income,  
    SUM(Person_home_ownership IS NULL) AS null_home_ownership,  
    SUM(Person_emp_length IS NULL) AS null_emp_length,  
    SUM(Loan_intent IS NULL) AS null_loan_intent,  
    SUM(Loan_grade IS NULL) AS null_loan_grade,  
    SUM(Loan_amnt IS NULL) AS null_loan_amount,  
    SUM(Loan_int_rate IS NULL) AS null_interest_rate,  
    SUM(Loan_status IS NULL) AS null_loan_status,  
    SUM(Loan_percent_income IS NULL) AS null_loan_percent_income,  
    SUM(CB_person_default_on_file IS NULL) AS null_previous_default,
```

```

SUM(CB_person_cred_hist_length IS NULL) AS null_credit_history
FROM credit_risk;

```

null_age	null_income	null_home_ownership	null_emp_length	null_loan_intent	null_loan_grade	null_loan_amount	null_interest_rate	null_loan_status	null_loan_percent_income
0	0	0	895	0	0	0	3116	0	0

null_previous_default	null_credit_history
0	0

Missing interest rate:

```

SELECT
CASE
WHEN Loan_int_rate IS NULL THEN 'Missing'
ELSE 'Available'
END AS interest_rate_status,
COUNT(*) AS total_loans,
SUM(Loan_status) AS defaulted_loans,
ROUND(100.0 * SUM(Loan_status) / COUNT(*),2) AS default_rate_pct
FROM credit_risk
GROUP BY interest_rate_status;

```

Result Grid				
Filter Rows:				
Export:				
Wrap Cell Content:				
	interest_rate_status	total_loans	defaulted_loans	default_rate_pct
▶	Available	29465	6464	21.94
	Missing	3116	644	20.67

Basic statistics:

```

SELECT COUNT(*) AS Total_loans,
SUM(Loan_amnt) AS Total_loan_amount,
AVG(Loan_amnt) AS Average_loan_amount,
MIN(Loan_amnt) AS Min_loan_amount,
MAX(Loan_amnt) AS Max_loan_amount,

```

```

AVG(Person_income) AS Average_income,
AVG(Loan_int_rate) AS Average_intrest_rate
FROM credit_risk;

```

Result Grid Filter Rows: Export: Wrap Cell Content:							
	Total_loans	Total_loan_amount	Average_loan_amount	Min_loan_amount	Max_loan_amount	Average_income	Average_intrest_rate
▶	32581	312431300	9589.3711	500	35000	66074.8485	11.01169489224439

CLEANING:

1) (Loan_status = 1 → Default , Loan_status = 0 → Non-default)

```

WITH loan_data AS (SELECT *,
CASE
WHEN Loan_status = 1 THEN 1 ELSE 0
END AS default_flag
FROM credit_risk
)
SELECT * FROM loan_data
LIMIT 10;

```

Person_emp_length	Loan_intent	Loan_grade	Loan_amnt	Loan_int_rate	Loan_status	Loan_percent_income	CB_person_default_on_file	CB_person_cred_hist_length	default_flag
123	PERSONAL	D	35000	16.02	1	0.59	Y	3	1
5	EDUCATION	B	1000	11.14	0	0.1	N	2	0
1	MEDICAL	C	5500	12.87	1	0.57	N	3	1
4	MEDICAL	C	35000	15.23	1	0.53	N	2	1
8	MEDICAL	C	35000	14.27	1	0.55	Y	4	1
2	VENTURE	A	2500	7.14	1	0.25	N	2	1
8	EDUCATION	B	35000	12.42	1	0.45	N	3	1
5	MEDICAL	B	35000	11.11	1	0.44	N	4	1
8	PERSONAL	A	35000	8.9	1	0.42	N	2	1
6	VENTURE	D	1600	14.74	1	0.16	N	3	1

2) Overall Default Rate:

```

SELECT COUNT(*) AS Total_loans,
SUM(CASE
WHEN Loan_status=1 THEN 1 ELSE 0
END) AS Defaulted_loans,
ROUND(100* SUM(CASE

```

```

WHEN Loan_status=1 THEN 1 ELSE 0 END)/COUNT(*),2) AS Default_rate_pct
FROM credit_risk;

```

Result Grid	Filter Rows:	Export:	Wrap Content
	Total_loans	Defaulted_loans	Default_rate_pct
▶	32581	7108	21.82

3) Default vs non-default portfolio:

```

SELECT
CASE
WHEN Loan_status=1 THEN 'Default' ELSE 'Non-Default' END AS Loan_result,

COUNT(*) AS Loan_result,

SUM(Loan_amnt) AS Total_loan_amount,

ROUND(100*COUNT(*)/SUM(COUNT(*)) OVER(),2) AS Portfolio_share_pct
FROM credit_risk
GROUP BY Loan_status;

```

Result Grid		Filter Rows		Export	Wrap Cell Content
	Loan_result	Loan_result	Total_loan_amount	Portfolio_share_pct	
▶	Default	7108	77125375	21.82	
	Non-Default	25473	235305925	78.18	

Handling NULL in person employment length:

```

SELECT
CASE
WHEN Person_emp_length IS NULL THEN 'Unknown'
WHEN Person_emp_length < 2 THEN '0-1 Years'
WHEN Person_emp_length < 5 THEN '2-4 Years'
WHEN Person_emp_length < 10 THEN '5-9 Years'
ELSE '10+ Years'
END AS employment_band,
COUNT(*) AS total_loans,
SUM(Loan_status) AS defaulted_loans,
ROUND(100.0 * SUM(Loan_status) / COUNT(*),2) AS default_rate_pct
FROM credit_risk
GROUP BY

```

```

CASE
WHEN Person_emp_length IS NULL THEN 'Unknown'
WHEN Person_emp_length < 2 THEN '0-1 Years'
WHEN Person_emp_length < 5 THEN '2-4 Years'
WHEN Person_emp_length < 10 THEN '5-9 Years'
ELSE '10+ Years'
END
ORDER BY default_rate_pct DESC;

```

Result Grid Filter Rows: Export: Wrap Cell Content: IA				
	employment_band	total_loans	defaulted_loans	default_rate_pct
▶	Unknown	895	282	31.51
	0-1 Years	7020	1953	27.82
	2-4 Years	10179	2293	22.53
	5-9 Years	10862	1985	18.27
	10+ Years	3625	595	16.41

Handling NULL in interest rate:

```

SELECT
    Loan_grade,
    COUNT(*) AS total_loans,
    SUM(Loan_int_rate IS NULL) AS missing_interest_rate,
    ROUND(100.0 * SUM(Loan_int_rate IS NULL) / COUNT(*),2) AS missing_pct
FROM credit_risk
GROUP BY Loan_grade
ORDER BY missing_interest_rate DESC;

```

Result Grid Filter Rows: Export: Wrap Cell Co				
	Loan_grade	total_loans	missing_interest_rate	missing_pct
	B	10451	1056	10.10
	A	10777	1003	9.31
	C	6458	630	9.76
	D	3626	312	8.60
	E	964	83	8.61
	F	241	27	11.20
	G	64	5	7.81

ANALYSIS

1) WHERE IS THE RISK?

1) Default rate by Loan Intent:

```
SELECT Loan_intent,  
COUNT(*) AS Total_loans,  
  
SUM(Loan_status) AS Defaulted_loan,  
  
ROUND(100*SUM(Loan_status)/count(*),2) AS Default_rate_pct,  
  
SUM(Loan_amnt) AS Total_loan_amount,  
  
ROUND(AVG(Loan_amnt),2) AS Avg_loan_amount  
  
FROM credit_risk  
GROUP BY Loan_intent  
ORDER BY Default_rate_pct DESC;
```



Loan_intent	Total_loans	Defaulted_loan	Default_rate_pct	Total_loan_amount	Avg_loan_amount
DEBTCONSOLIDATION	5212	1490	28.59	50008550	9594.89
MEDICAL	6071	1621	26.70	56214925	9259.58
HOMEIMPROVEMENT	3605	941	26.10	37349675	10360.52
PERSONAL	5521	1098	19.89	52856800	9573.77
EDUCATION	6453	1111	17.22	61191725	9482.68
VENTURE	5719	847	14.81	54809625	9583.78

2) Default rate by Person_home_ownership:

```
SELECT Person_home_ownership,  
  
COUNT(*) AS Total_loans,
```

SUM(Loan_status) AS Defaulted_loan,

ROUND(100*SUM(Loan_status)/count(*),2) AS Default_rate_pct,

SUM(Loan_amnt) AS Total_loan_amount,

ROUND(AVG(Loan_amnt),2) AS Avg_loan_amount

FROM credit_risk

GROUP BY Person_home_ownership

ORDER BY Default_rate_pct DESC;

	Person_home_ownership	Total_loans	Defaulted_loan	Default_rate_pct	Total_loan_amount	Avg_loan_amount
►	RENT	16446	5192	31.57	145749900	8862.33
	OTHER	107	33	30.84	1184975	11074.53
	MORTGAGE	13444	1690	12.57	142163050	10574.46
	OWN	2584	193	7.47	23333375	9029.94

3) Default rate by Loan Grade:

SELECT Loan_grade,

COUNT(*) AS Total_loans,

SUM(Loan_status) AS Defaulted_loan,

ROUND(100*SUM(Loan_status)/count(*),2) AS Default_rate_pct,

SUM(Loan_amnt) AS Total_loan_amount,

ROUND(AVG(Loan_amnt),2) AS Avg_loan_amount

FROM credit_risk

GROUP BY Loan_grade

ORDER BY Default_rate_pct DESC;

Loan_grade	Total_loans	Defaulted_loan	Default_rate_pct	Total_loan_amount	Avg_loan_amount
G	64	63	98.44	1100525	17195.70
F	241	170	70.54	3546875	14717.32
E	964	621	64.42	12450875	12915.85
D	3626	2141	59.05	39339350	10849.24
C	6458	1339	20.73	59503125	9213.86
B	10451	1701	16.28	104462800	9995.48
A	10777	1073	9.96	92027750	8539.27

4) Default rate by previous credit default:

SELECT CB_person_default_on_file,
COUNT(*) AS Total_loans,

SUM(Loan_status) AS Defaulted_loan,

ROUND(100*SUM(Loan_status)/count(*),2) AS Default_rate_pct,

SUM(Loan_amnt) AS Total_loan_amount,

ROUND(AVG(Loan_amnt),2) AS Avg_loan_amount

FROM credit_risk

GROUP BY CB_person_default_on_file

ORDER BY Default_rate_pct DESC;

	CB_person_default_on_file	Total_loans	Defaulted_loan	Default_rate_pct	Total_loan_amount	Avg_loan_amount
Y		5745	2172	37.81	58158700	10123.36
N		26836	4936	18.39	254272600	9475.06

5) Default rate by age groups:

```

WITH age_groups AS(
SELECT*,
CASE
WHEN Person_age <25 THEN 'Under 25'
WHEN Person_age BETWEEN 25 AND 34 THEN '25-34'
WHEN Person_age BETWEEN 34 AND 44 THEN '34-44'
WHEN Person_age BETWEEN 44 AND 54 THEN '44-54'
ELSE '55+'
END AS age_group
FROM credit_risk)
SELECT age_group,
COUNT(*) AS Total_loans,
SUM(Loan_status) AS Defaulted_loan,
ROUND(100*SUM(Loan_status)/count(*),2) AS Default_rate_pct,
SUM(Loan_amnt) AS Total_loan_amount,
ROUND(AVG(Loan_amnt),2) AS Avg_loan_amount
FROM age_groups
GROUP BY age_group
ORDER BY Default_rate_pct DESC;

```

age_group	Total_loans	Defaulted_loan	Default_rate_pct	Total_loan_amount	Avg_loan_amount
55+	159	39	24.53	1641825	10325.94
Under 25	12315	2860	23.22	111681950	9068.77
44-54	601	132	21.96	5985975	9960.02
25-34	16180	3393	20.97	160056900	9892.27
34-44	3326	684	20.57	33064650	9941.27

6) Default rate by Income groups:

```

WITH Income_groups AS(
SELECT*,

```

```

CASE
WHEN Person_income <30000 THEN 'low income'
WHEN Person_income <60000 THEN 'lower-middle income'
WHEN Person_income <100000 THEN 'middle income'
ELSE 'high income'
END AS income_group
FROM credit_risk)
SELECT income_group,
COUNT(*) AS Total_loans,
SUM(Loan_status) AS Defaulted_loan,
ROUND(100*SUM(Loan_status)/count(*),2) AS Default_rate_pct,
SUM(Loan_amnt) AS Total_loan_amount,
ROUND(AVG(Loan_amnt),2) AS Avg_loan_amount
FROM Income_groups
GROUP BY income_group
ORDER BY Default_rate_pct DESC;

```

	income_group	Total_loans	Defaulted_loan	Default_rate_pct	Total_loan_amount	Avg_loan_amount
►	low income	3669	1727	47.07	18894675	5149.82
	lower-middle income	14011	3614	25.79	115205025	8222.47
	middle income	10331	1359	13.15	113883700	11023.49
	high income	4570	408	8.93	64447900	14102.39

2) What factors are associated with default?

:

- 1) Default rate by Loan interest rate :

```

WITH interest_groups AS(
SELECT*,
CASE
WHEN Loan_int_rate IS NULL THEN 'Missing_rate'
WHEN Loan_int_rate < 8 THEN 'Below 8%'
WHEN Loan_int_rate < 12 THEN '8%-12%'
WHEN Loan_int_rate < 16 THEN '12%-16%'
ELSE '16+'
END AS interest_rate_group
FROM credit_risk)
SELECT interest_rate_group,
COUNT(*) AS Toatal_loans,
SUM(Loan_status) AS Defaulted_loan,
ROUND(100*SUM(Loan_status)/COUNT(*),2) AS Default_rate_pct
FROM interest_groups
GROUP BY interest_rate_group
ORDER BY Default_rate_pct DESC;

```

Result Grid Filter Rows: Export: Wrap Cell Content:				
	interest_rate_group	Toatal_loans	Defaulted_loan	Default_rate_pct
▶	16+	1875	1186	63.25
	12%-16%	9272	2896	31.23
	Missing_rate	3116	644	20.67
	8%-12%	10548	1652	15.66
	Below 8%	7770	730	9.40

2) Default rate by Loan Amount:

```

WITH Loan_amount_groups AS(
SELECT*,

```

```

CASE
WHEN Loan_amnt < 5000 THEN 'Below 5k'
WHEN Loan_amnt < 10000 THEN '5k-10k'
WHEN Loan_amnt < 20000 THEN '10k-20k'
ELSE '20k+'
END AS Loan_amount_group
FROM credit_risk)

SELECT Loan_amount_group,
COUNT(*) AS Total_loans,
SUM(Loan_status) AS Defaulted_loan,
ROUND(100*SUM(Loan_status)/COUNT(*),2) AS Default_rate_pct
FROM Loan_amount_groups
GROUP BY Loan_amount_group
ORDER BY Default_rate_pct DESC;

```

Result Grid Filter Rows: Export: Wrap Cell Content: IA				
	Loan_amount_group	Total_loans	Defaulted_loan	Default_rate_pct
▶	20k+	3156	1063	33.68
	10k-20k	10560	2447	23.17
	Below 5k	7446	1547	20.78
	5k-10k	11419	2051	17.96

3) Default rate by Person Employment length:

```

WITH Employment_groups AS(
SELECT*,
CASE
WHEN Person_emp_length IS NULL THEN 'Unkown'
WHEN Person_emp_length < 2 THEN '0-1 years'

```

```

WHEN Person_emp_length < 5 THEN '2-5 years'
WHEN Person_emp_length < 10 THEN '5-10 years'
ELSE '10+ years'
END AS Employment_group
FROM credit_risk)
SELECT Employment_group,
COUNT(*) AS Total_loans,
SUM(Loan_status) AS Defaulted_loan,
ROUND(100*SUM(Loan_status)/COUNT(*),2) AS Default_rate_pct
FROM Employment_groups
GROUP BY Employment_group
ORDER BY Default_rate_pct DESC;

```

Employment_group	Total_loans	Defaulted_loan	Default_rate_pct
Unkown	895	282	31.51
0-1 years	7020	1953	27.82
2-5 years	10179	2293	22.53
5-10 years	10862	1985	18.27
10+ years	3625	595	16.41

4) Default Rate by Credit history length:

```

WITH Credit_history_groups AS(
SELECT*,
CASE
WHEN CB_person_cred_hist_length < 5 THEN '0-5 years'
WHEN CB_person_cred_hist_length < 10 THEN '5-10 years'
WHEN CB_person_cred_hist_length < 15 THEN '10-15 years'
ELSE '15+ years'
END AS Credit_history_group
FROM credit_risk)
SELECT Credit_history_group,
COUNT(*) AS Total_loans,

```

```

SUM(Loan_status) AS Defaulted_loan,
ROUND(100*SUM(Loan_status)/COUNT(*),2) AS Default_rate_pct
FROM Credit_history_groups
GROUP BY Credit_history_group
ORDER BY Default_rate_pct DESC;

```

	Credit_history_group	Total_loans	Defaulted_loan	Default_rate_pct
▶	0-5 years	17833	4052	22.72
	15+ years	1579	343	21.72
	5-10 years	9436	1954	20.71
	10-15 years	3733	759	20.33

5) Default Rate Loan percent income:

```

WITH Income_burden AS(
SELECT*,
CASE
WHEN Loan_percent_income < 0.10 THEN 'Below 10%'
WHEN Loan_percent_income < 0.20 THEN '10%-20%'
WHEN Loan_percent_income < 0.30 THEN '20%-30%'
ELSE '30%+'
END AS Income_burden_group
FROM credit_risk)
SELECT Income_burden_group,
COUNT(*) AS Total_loans,
SUM(Loan_status) AS Defaulted_loan,
ROUND(100*SUM(Loan_status)/COUNT(*),2) AS Default_rate_pct
FROM Income_burden
GROUP BY Income_burden_group
ORDER BY Default_rate_pct DESC;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Income_burden_group	Total_loans	Defaulted_loan	Default_rate_pct
	30%+	4285	2805	65.46
	20%-30%	6817	1424	20.89
	10%-20%	12528	1846	14.73
	Below 10%	8951	1033	11.54

6). Compare average characteristics between defaulted and non-defaulted borrowers.

SELECT

CASE

WHEN Loan_status = 1 THEN 'Default'

ELSE 'Non-Default'

END AS loan_result,

COUNT(*) AS total_loans,

ROUND(AVG(Person_income), 2) AS avg_income,

ROUND(AVG(Loan_amnt), 2) AS avg_loan_amount,

ROUND(AVG(Loan_int_rate), 2) AS avg_interest_rate,

ROUND(AVG(Person_emp_length), 2) AS avg_employment_length,

ROUND(AVG(CB_person_cred_hist_length), 2) AS avg_credit_history,

ROUND(AVG(Loan_percent_income), 3) AS avg_loan_income_ratio

FROM credit_risk

GROUP BY Loan_status;

Result Grid | Filter Rows: | Export: | Wrap Cell Content: ☐

	loan_result	total_loans	avg_income	avg_loan_amount	avg_interest_rate	avg_employment_length	avg_credit_history	avg_loan_income_ratio
▶	Default	7108	49125.65	10850.50	13.06	4.14	5.69	0.247
	Non-Default	25473	70804.36	9237.46	10.44	4.97	5.84	0.149

7. Comparison between Loan grade + previous default:

```

SELECT Loan_grade,CB_person_default_on_file AS previous_default,
COUNT(*) AS Total_loans,
SUM(Loan_status) AS Default_loans,
ROUND(100.0*SUM(Loan_status)/COUNT(*),2) AS Default_rate_pct,
SUM(Loan_amnt) AS Total_loan_amount
FROM credit_risk
GROUP BY Loan_grade,CB_person_default_on_file
HAVING COUNT(*) >= 50
ORDER BY Default_rate_pct DESC;

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 						
	Loan_grade	previous_default	Total_loans	Default_loans	Default_rate_pct	Total_loan_amount
▶	F	N	129	92	71.32	1973475
	F	Y	112	78	69.64	1573400
	E	N	499	330	66.13	6447975
	E	Y	465	291	62.58	6002900
	D	Y	1876	1127	60.07	20556925
	D	N	1750	1014	57.94	18782425
	C	N	3202	698	21.80	30104700
	C	Y	3256	641	19.69	29398425
	B	N	10451	1701	16.28	104462800
	A	N	10777	1073	9.96	92027750

8. Comparison between Loan intent + loan grade:

```

SELECT Loan_grade,Loan_intent,
COUNT(*) AS Total_loans,
SUM(Loan_status) AS Default_loans,
ROUND(100.0*SUM(Loan_status)/COUNT(*),2) AS Default_rate_pct,
SUM(Loan_amnt) AS Total_loan_amount
FROM credit_risk
GROUP BY Loan_grade,Loan_intent
HAVING COUNT(*) >= 50

```

ORDER BY Default_rate_pct DESC;

Loan_grade	Loan_intent	Total_loans	Default_loans	Default_rate_pct	Total_loan_am
E	DEBTCONSOLIDATION	144	144	100.00	1810950
F	MEDICAL	52	50	96.15	746500
E	MEDICAL	167	160	95.81	2158625
D	DEBTCONSOLIDATION	600	556	92.67	6430575
D	MEDICAL	658	577	87.69	6995175
D	HOMEIMPROVEMENT	487	263	54.00	5546300
E	HOMEIMPROVEMENT	143	74	51.75	2022725
E	VENTURE	178	92	51.69	2251625
E	PERSONAL	147	71	48.30	1845400
D	PERSONAL	643	280	43.55	6998750
E	EDUCATION	185	80	43.24	2361550
D	VENTURE	581	233	40.10	6215975
D	EDUCATION	657	232	35.31	7152575
C	HOMEIMPROVEMENT	693	189	27.27	6954600
C	DEBTCONSOLIDATION	1042	237	22.74	9534525

B	HOMEIMPROVEMENT	1115	240	21.52	12083900
C	MEDICAL	1217	256	21.04	10864875
C	VENTURE	1104	215	19.47	10151400
C	PERSONAL	1046	198	18.93	9933275
B	DEBTCONSOLIDATION	1620	303	18.70	16432775
B	PERSONAL	1813	330	18.20	17683125
C	EDUCATION	1356	244	17.99	12064450
B	MEDICAL	2006	341	17.00	19005075
B	EDUCATION	2026	317	15.65	20318300
A	HOMEIMPROVEMENT	1126	145	12.88	10006250
A	MEDICAL	1959	225	11.49	16261675
A	DEBTCONSOLIDATION	1753	197	11.24	14977750
A	PERSONAL	1832	194	10.59	15758325
A	EDUCATION	2174	202	9.29	18472075
B	VENTURE	1871	170	9.09	18939625

A	VENTURE	1933	110	5.69	16551675
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3 Where is the portfolio exposure?

1) Portfolio exposure by loan grade:

SELECT Loan_grade,

COUNT(*) AS total_loans,

```

SUM(Loan_amnt) AS total_exposure,
SUM(
CASE
    WHEN Loan_status = 1 THEN Loan_amnt ELSE 0 END) AS defaulted_exposure,
ROUND(100.0 * SUM(Loan_status) / COUNT(*),2) AS default_rate_pct
FROM credit_risk
GROUP BY Loan_grade
ORDER BY defaulted_exposure DESC;

```

	Loan_grade	total_loans	total_exposure	defaulted_exposure	default_rate_pct
▶	D	3626	39339350	22800100	59.05
	B	10451	104462800	19120425	16.28
	C	6458	59503125	13579600	20.73
	A	10777	92027750	10202525	9.96
	E	964	12450875	7826925	64.42
	F	241	3546875	2496875	70.54
	G	64	1100525	1098925	98.44

2) Portfolio exposure by loan intent:

```

SELECT Loan_intent,
COUNT(*) AS total_loans,
SUM(Loan_amnt) AS total_exposure,
SUM(
CASE
    WHEN Loan_status = 1 THEN Loan_amnt ELSE 0 END) AS defaulted_exposure,
ROUND(100.0 * SUM(Loan_status) / COUNT(*),2) AS default_rate_pct
FROM credit_risk
GROUP BY Loan_intent
ORDER BY defaulted_exposure DESC;

```

Loan_intent	total_loans	total_exposure	defaulted_exposure	default_rate_pct
MEDICAL	6071	56214925	18146000	26.70
DEBTCONSOLIDATION	5212	50008550	16561650	28.59
EDUCATION	6453	61191725	12156450	17.22
PERSONAL	5521	52856800	11542800	19.89
VENTURE	5719	54809625	9412600	14.81
HOMEIMPROVEMENT	3605	37349675	9305875	26.10

3) Portfolio exposure by home ownership

```

SELECT Person_home_ownership,
       COUNT(*) AS total_loans,
       SUM(Loan_amnt) AS total_exposure,
       SUM(
         CASE
           WHEN Loan_status = 1 THEN Loan_amnt ELSE 0 END) AS defaulted_exposure,
       ROUND(100.0 * SUM(Loan_status) / COUNT(*),2) AS default_rate_pct
FROM credit_risk
GROUP BY Person_home_ownership
ORDER BY defaulted_exposure DESC;

```

Person_home_ownership	total_loans	total_exposure	defaulted_exposure	default_rate_pct
RENT	16446	145749900	55865775	31.57
MORTGAGE	13444	142163050	19519350	12.57
OWN	2584	23333375	1298175	7.47
OTHER	107	1184975	442075	30.84

RISK + EXPOSURE MATRIX

4) Classify segments according to: Default rate Portfolio exposure Thresholds

```

WITH segment_risk AS (
SELECT
  Loan_grade,

```

```
COUNT(*) AS total_loans,  
SUM(Loan_amnt) AS total_exposure,  
SUM(  
CASE  
WHEN Loan_status = 1 THEN Loan_amnt ELSE 0 END) AS defaulted_exposure,  
100.0 * SUM(Loan_status) / COUNT(*) AS default_rate_pct
```

```
FROM credit_risk  
GROUP BY Loan_grade),
```

```
portfolio_average AS (  
SELECT  
AVG(default_rate_pct) AS avg_default_rate,  
AVG(total_exposure) AS avg_exposure  
FROM segment_risk)
```

```
SELECT  
s.Loan_grade,  
s.total_loans,  
ROUND(s.default_rate_pct, 2) AS default_rate_pct,  
s.total_exposure,  
s.defaulted_exposure,  
CASE  
WHEN s.default_rate_pct >= p.avg_default_rate AND s.total_exposure >= p.avg_exposure  
THEN 'High Risk + High Exposure'  
WHEN s.default_rate_pct >= p.avg_default_rate AND s.total_exposure < p.avg_exposure  
THEN 'High Risk + Low Exposure'
```

```

WHEN s.default_rate_pct < p.avg_default_rate AND s.total_exposure >= p.avg_exposure
THEN 'Low Risk + High Exposure'

```

```

ELSE 'Low Risk + Low Exposure'

```

```

END AS risk_exposure_category

```

```

FROM segment_risk s

```

```

CROSS JOIN portfolio_average p

```

```

ORDER BY s.default_rate_pct DESC;

```

	Loan_grade	total_loans	default_rate_pct	total_exposure	defaulted_exposure	risk_exposure_category
▶	G	64	98.44	1100525	1098925	High Risk + Low Exposure
	F	241	70.54	3546875	2496875	High Risk + Low Exposure
	E	964	64.42	12450875	7826925	High Risk + Low Exposure
	D	3626	59.05	39339350	22800100	High Risk + Low Exposure
	C	6458	20.73	59503125	13579600	Low Risk + High Exposure
	B	10451	16.28	104462800	19120425	Low Risk + High Exposure
	A	10777	9.96	92027750	10202525	Low Risk + High Exposure

5) RISK + EXPOSURE RANKING BY LOAN GRADE

```

WITH segment_analysis AS (

```

```

SELECT

```

```

Loan_grade,

```

```

COUNT(*) AS total_loans,

```

```

SUM(Loan_status) AS defaulted_loans,

```

```

ROUND(100.0 * SUM(Loan_status) / COUNT(*), 2) AS default_rate_pct,

```

```

SUM(Loan_amnt) AS total_exposure,

```

```

SUM(

```

```

CASE

```

```
WHEN Loan_status = 1 THEN Loan_amnt ELSE 0 END) AS defaulted_exposure
```

```
FROM credit_risk
```

```
GROUP BY Loan_grade),
```

```
ranked_segments AS (
```

```
SELECT*,
```

```
RANK() OVER (ORDER BY default_rate_pct DESC) AS default_rate_rank,
```

```
RANK() OVER (ORDER BY defaulted_exposure DESC) AS defaulted_exposure_rank
```

```
FROM segment_analysis)
```

```
SELECT
```

```
    Loan_grade,
```

```
    total_loans,
```

```
    defaulted_loans,
```

```
    default_rate_pct,
```

```
    total_exposure,
```

```
    defaulted_exposure,
```

```
    default_rate_rank,
```

```
    defaulted_exposure_rank,
```

```
CASE
```

```
WHEN default_rate_rank = 1 AND defaulted_exposure_rank = 1 THEN 'Highest Risk +  
Highest Exposure'
```

```
WHEN default_rate_rank = 1 THEN 'Highest Default Rate'
```

```
WHEN defaulted_exposure_rank = 1 THEN 'Highest Defaulted Exposure'
```

```
ELSE 'Other'
```

```
END AS risk_priority
```

FROM ranked_segments

ORDER BY default_rate_rank;

Loan_grade	total_loans	defaulted_loans	default_rate_pct	total_exposure	defaulted_exposure	default_rate_rank	defaulted_exposure_rank	risk_priority
G	64	63	98.44	1100525	1098925	1	7	Highest Default Rate
F	241	170	70.54	3546875	2496875	2	6	Other
E	964	621	64.42	12450875	7826925	3	5	Other
D	3626	2141	59.05	39339350	22800100	4	1	Highest Defaulted Exposure
C	6458	1339	20.73	59503125	13579600	5	3	Other
B	10451	1701	16.28	104462800	19120425	6	2	Other
A	10777	1073	9.96	92027750	10202525	7	4	Other

6) HIGH-RISK LOAN INTENT + LOAN GRADE COMBINATIONS

WITH segment_analysis AS (

SELECT

Loan_intent,

Loan_grade,

COUNT(*) AS total_loans,

SUM(Loan_status) AS defaulted_loans,

ROUND(100.0 * SUM(Loan_status) / COUNT(*),2) AS default_rate_pct,

SUM(Loan_amnt) AS total_exposure,

SUM(

CASE

WHEN Loan_status = 1 THEN Loan_amnt ELSE 0 END) AS defaulted_exposure

FROM credit_risk

GROUP BY Loan_intent,Loan_grade

HAVING COUNT(*) >= 50

),

ranked_segments AS (

SELECT

*,

```

RANK() OVER (
    ORDER BY default_rate_pct DESC
) AS risk_rank
FROM segment_analysis)
SELECT
    Loan_intent,
    Loan_grade,
    total_loans,
    defaulted_loans,
    default_rate_pct,
    total_exposure,
    defaulted_exposure,
    risk_rank
FROM ranked_segments
WHERE risk_rank <= 10
ORDER BY risk_rank, defaulted_exposure DESC;

```

Result Grid Filter Rows: Export: Wrap Cell Content:								
Loan_intent	Loan_grade	total_loans	defaulted_loans	default_rate_pct	total_exposure	defaulted_exposure	risk_rank	
DEBTCONSOLIDATION	E	144	144	100.00	1810950	1810950	1	
MEDICAL	F	52	50	96.15	746500	743800	2	
MEDICAL	E	167	160	95.81	2158625	2101675	3	
DEBTCONSOLIDATION	D	600	556	92.67	6430575	6071375	4	
MEDICAL	D	658	577	87.69	6995175	6260100	5	
HOMEIMPROVEMENT	D	487	263	54.00	5546300	2639525	6	
HOMEIMPROVEMENT	E	143	74	51.75	2022725	895250	7	
VENTURE	E	178	92	51.69	2251625	1115025	8	
PERSONAL	E	147	71	48.30	1845400	835600	9	
PERSONAL	D	643	280	43.55	6998750	2903375	10	

7) DEFAULT RATE VS PORTFOLIO SHARE

"Is the segment risky AND large enough to matter?"

```
WITH segment_summary AS (
```

```
SELECT
```

```
Loan_intent,
```

```
COUNT(*) AS total_loans,
```

```
SUM(Loan_amnt) AS total_exposure,
```

```
SUM(Loan_status) AS defaulted_loans,
```

```
100.0 * SUM(Loan_status) / COUNT(*) AS default_rate_pct
```

```
FROM credit_risk
```

```
GROUP BY Loan_intent)
```

```
SELECT
```

```
Loan_intent,
```

```
total_loans,
```

```
defaulted_loans,
```

```
ROUND(default_rate_pct, 2) AS default_rate_pct,
```

```
total_exposure,
```

```
ROUND(100.0 * total_exposure / SUM(total_exposure) OVER (), 2) AS portfolio_exposure_pct
```

```
FROM segment_summary
```

```
ORDER BY default_rate_pct DESC;
```

Result Grid		Filter Rows:		Export:	Wrap Cell Content:	
	Loan_intent	total_loans	defaulted_loans	default_rate_pct	total_exposure	portfolio_exposure_pct
►	DEBTCONSOLIDATION	5212	1490	28.59	50008550	16.01
	MEDICAL	6071	1621	26.70	56214925	17.99
	HOMEIMPROVEMENT	3605	941	26.10	37349675	11.95
	PERSONAL	5521	1098	19.89	52856800	16.92
	EDUCATION	6453	1111	17.22	61191725	19.59
	VENTURE	5719	847	14.81	54809625	17.54